



PROFILE SUMMARY

Researcher in formal reasoning with a background in mathematics & computer science. My current research focuses on automated reasoning, SMT solving, and formal verification, particularly satisfiability certification for quantified formulas. More broadly, I am interested in the connections between mathematical logic, theorem proving, and automated methods for solving computational problems.

CONTACT DETAILS

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RESEARCH INTEREST

- Formal Verification
- Automated Reasoning
- Theorem Proving
- Logical Foundations of Computation
- Formalised Mathematics
- Foundations of Mathematics
- Philosophical logic

TECHNICAL SKILLS

- Language: Bahasa Indonesia, English
- Programming: Python (developing), Java, C
- Scientific Computing: Maple, Matlab
- Typesetting: LaTeX

EXPERIENCE

VĚDECKÝ ASISTENT (SENIOR RESEARCH ASSISTANT)
*Institute of Computer Science,
Czech Academy of Sciences.* **08.09.2025–present**

◊ Member of the Department of Artificial Intelligence. Research involving automated reasoning, SMT solvers, and AI methods. Currently pursuing a second PhD in related areas.

LECTURER (ASSISTANT PROFESSORIAL FELLOW 5)
*Institute of Mathematics
The University of Philippines–Diliman* **01.08.2024–31.07.2025**

◊ Deliver undergraduate lectures and exams and prepare teaching materials.

CAREER INTERRUPTION **2019–2024**
◊ Academic career break devoted to religious vocation & monastic formation.

TUTOR
*School of Mathematics & Statistics
University of Canterbury, NZ* **2015–2018**
◊ Leading tutorial classes (about 20 students in one class), marking exams, proctor for quizzes and exams.

LECTURER AND RESEARCHER
*Faculty of Computer Science
University of Indonesia* **2012–2014**
◊ Prepare and give classes on the subjects assigned by the Dean, make and mark exams, manage some teaching assistants, member of Formal Methods in Software Engineering lab.

EDUCATION

PHD IN COMPUTER SCIENCE.
Czech Technical University (ČVUT). **2025–present**

PHD IN MATHEMATICS.
University of Canterbury (NZ). **2014–2019**

MASTER OF COMPUTER SCIENCE.
University of Indonesia. **2011–2012**

BACHELOR OF SCIENCE IN MATHEMATICS
University of Indonesia. **2006–2009**

PUBLICATIONS

1. Anggha Nugraha. *Paraconsistent Dominated Convergence*. In arXiv:2607.15723 [math.LO], 2026.
2. Anggha Nugraha. *A Constructive Field of Infinitesimals: Chunk and Permeate Approach*.¹ In arXiv:2607.08206 [math.LO], 2026.
3. Stefan Ratschan, Anggha Nugraha, Mikoláš Janota, Marek Dančo. *Inductive Satisfiability Certification for Universal Quantifiers and Uninterpreted Function Symbols*. In arXiv:2602.16335 [cs.LO], 2026.
4. Anggha Nugraha, M. McKubre-Jordens, and H. Diener. *Naïve Infinitesimal Analysis: Its Construction and Its Properties*. In arXiv.2009.11424 [math.LO], 2020.

5. T. Antaresti, Anggha Nugraha, et al. *Wavelet denoising and fractal feature selection for classifying simulated earthquake signals from mobile phone accelerometers*. In IEEE 11th International Multi-Conference on Systems, Signals & Devices (SSD 14), 2014.
6. M. Arzaki and Anggha S. Nugraha. *Translation of an Allen's Temporal Logic Variant to RPNL*. In Journal of Mathematical and Computational Science (3), 2013, pages 1405-1429.
7. A. S. Nugraha and T. Basaruddin. *Analysis and Comparison of QR Decomposition Algorithms for Some Types of Matrix*. In Proceedings of Federated Conference on Computer Science and Information System (FedCSIS), 2012, pages 561-565.

CONFERENCES AND TALKS

- ◊ New Zealand Mathematics and Statistics Postgraduate Conference. Auckland, New Zealand in 2014, 2017, and 2018
- ◊ Various informal talks in The Netherlands (especially at the University of Amsterdam) when doing research collaboration with Prof. Jan Bergstra in The Netherlands in 2016
- ◊ 7th Non-Classical Logic: Theory and Applications conference in Toruń, Poland in 2015
- ◊ 1st National Joint Seminar in Mathematics at the University of Indonesia in 2010, *The Generating Function Applications for Counting The Number of Pattern Graph from Main Pattern Graph*.

GRANTS AND SCHOLARSHIPS

- ◊ LPDP Scholarship from the Ministry of Finance of The Republic of Indonesia 2014-2019
Scholarship to do my PhD study at the University of Canterbury, NZ.
- ◊ The New Zealand Mathematical Society (NZMS) Travel Grant 2016
Travel grant to do research collaboration with Prof. Jan Bergstra of the University of Amsterdam, The Netherlands.
- ◊ The New Zealand Mathematical Society (NZMS) Travel Grant 2015
Travel grant to present my paper at "The 7th Conference on Non-Classical Logic: Theory and Application" in Poland.
- ◊ IMHERE (Indonesia Managing Higher Education for Relevance and Efficiency) Scholarship for Master Study 2011-2012

REFERENCES

1. doc. Dipl.Ing. Stefan Ratschan, Dr.-tech.
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2. Dr. Maarten McKubre-Jordens
School of Mathematics and Statistics University of Canterbury
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3. Prof. Dr. Kiki Ariyanti Sugeng
Department of Mathematics University of Indonesia
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4. Phillipa Gourdie (Senior Tutor)
School of Mathematics and Statistics University of Canterbury
phillipa.gourdie@canterbury.ac.nz